Imperial College
London

Software Engineering 1: Introduction to Computing

Assignment – Spring 2017

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Write a C++ program for the 2048 game (http://gabrielecirulli.github.io/2048/).

The following screenshots show the game interface and expand on more specific requirements.

Let file `inputconf.txt` contained in the same directory of the executable (this, as usual, assuming the program is run from the command line, conditions may vary if an IDE is used) have the following content:

```
0 0 16 4
32 0 4 8
128 2 2 2
0 0 0 2
```

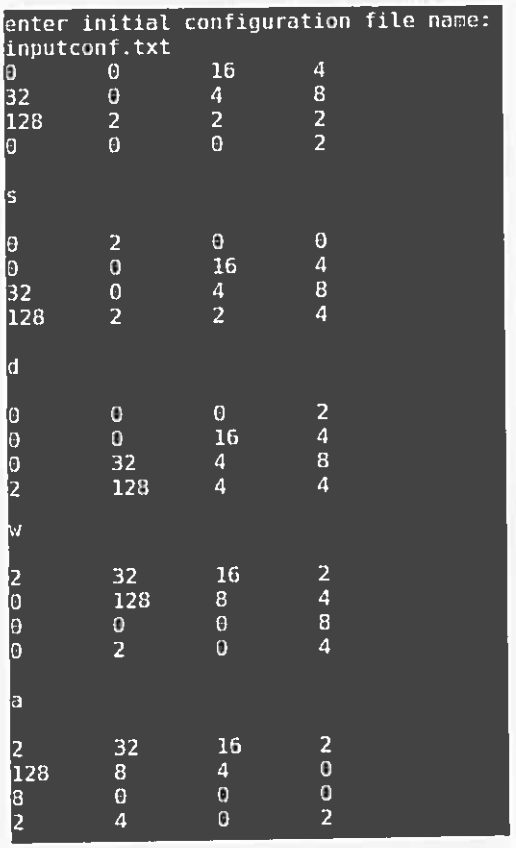


Figure 1: The program initiàlly asks for the name of a file containing the initial configuration. If the file is found, it is loaded and the game begins from said configuration. Commands are given in input using `a` for left, `s` for down, `d` for right and `w` for up. The letters are given in input in the usual way (i.e. followed by enter). In the original game either a 2 or a 4, with a certain probability, is placed randomly in one of the free tiles after a move. For this assignment it should always be a 2.

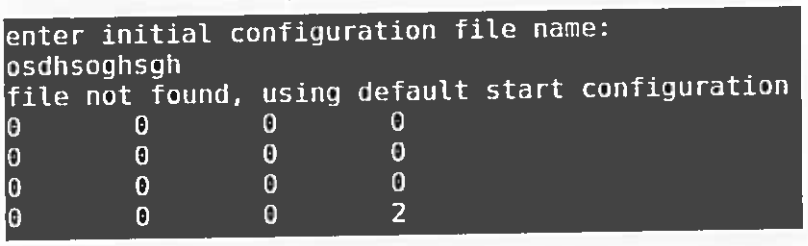


Figure 2: If the input file is not found then a message is printed and the game starts from a default configuration (a 2 in the bottom right corner).

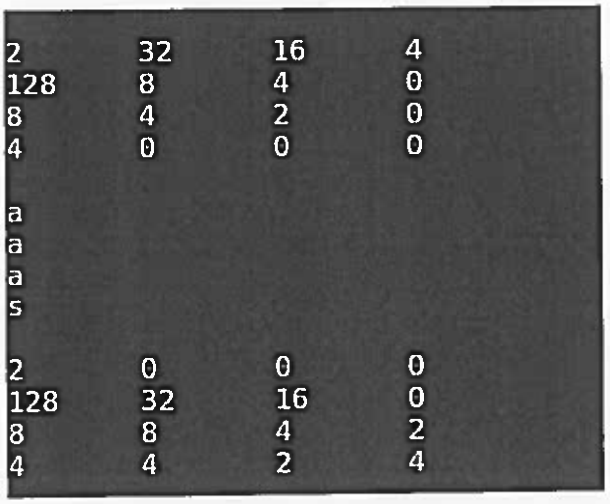


Figure 3: What should happen when a certain move is entered by the user but cannot be executed.



Figure 4: No possible moves: game over. This is the only termination condition: the game should go on even when the 2048 value is reached.

The lack of indications with respect to the structure of the program design and implementation is deliberate as one of the main assessment criteria for this assignment is about your ability to deal with these aspects autonomously (e.g. functional decomposition: identifying which functions should be defined, what should be in their implementation, their parameters etc).

Let me emphasize that bad design choices (e.g. everything implemented in the `main`) will be penalized accordingly.

Together with the source file (`.cpp`), you should submit a short summary in pdf format describing and motivating your design choices (e.g. which functions you defined and used and why you chose that structure for the program). Submit a zip archive containing both.

Functions other than the `main` should not contain user or file input or output (`cin`, `cout`, `ifstream` etc) in their implementation. However there is one exception: you may want to define and use a function whose (only) purpose is to print the "square" with the numbers; this is allowed and of course therefore the inclusion in this function of e.g. `cout` is allowed too.

All the variables should be declared in the scope of a function (either the `main` or some other one): there should be no global variables.

All the loops should be controlled either by the loop condition or by `return`: there should be no `break`, `continue` or `goto` statements anywhere in the program.

Functions should not *return* vectors provided as output, vectors should be provided in output using output parameters (i.e. passing by reference).

The use of the header `<algorithm>` is not allowed.

Only headers from the standard library are allowed.

Don't use C++11 features unless you are entirely sure of what you are doing. If you use C++11 features you must indicate this in a comment on top of the source file.

Do not use non-standard features. The program should be compiled and tested on Linux on the lab computers before the submission, if it can be compiled there, then it is alright.